

Technology #165

apply a magnetorheological (MR) fluid coating
to the rail surface and energize it with a
magnetic field during service to generate a
compressive stress field at potential crack
tips, thereby inducing a crack tip shielding
effect under cyclic loading

Engineering Technology-Transfer Dossier

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

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Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

apply a magnetorheological (MR) fluid coating to the rail surface and energize it with a magnetic field during service to generate a compressive stress field at potential crack tips, thereby inducing a crack tip shielding effect under cyclic loading

WHY DOES IT MATTER?

Documented problem 'fatigue fracture' concerning railway rail. User-submitted engineering problem (session input): "Why do rails fracture in service under fatigue loading?". [cod_optimade] retrieval status EMPTY — provider outcome recorded, not treated as absence (Art. XXI.3). [arxiv] 5 records retrieved live (2010-2020). Illustrative records: 'How we can control the crack to propagate along the specified path feasibly?'; 'Steady-state propagation of a Mode III crack in couple stress elastic materials'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 12 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in mechanical_structural. 10-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 1 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 10-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: embed strain gauges and acoustic emission sensors on a test rail segment with a seeded crack, run controlled tension■compression fatigue.... Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: buckling / kink margin tests across duty cycle. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: stress concentrators multiply local stress above the nominal value — geometry factors must be applied from the actual design; superelastic members (e.g. nitinol) do not follow linear elasticity through transformation — use their measured plateau relations instead; K for catheter constraint conditions is MODEL_DERIVED; MEC-002 (Section bending stress) is applicable only under: linear-elastic material; small deflection; prismatic section.; MEC-003 (Hooke's law (linear elasticity)) is applicable only under: below proportional limit; isothermal.; MEC-001 (Euler buckling load) is applicable only under: elastic, slender column, small deflection.; Model input variables requiring sourced values before any numeric result: M (N.m), c (m), I (m⁴), E (Pa), epsilon (-), E (Pa), I (m⁴), K (-), L (m) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: sigma (Pa), sigma (Pa), P_cr (N) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): member slenderness ratio [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; yield/ultimate strength [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; applied cyclic load amplitude [P_cr, N] value_status=UNKNOWN; required run-out cycles [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; articulation radius [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; friction coefficient at guides [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

1 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: crack tip shielding effect that correlates crack growth rate with the loading force range ΔF in addition to the stress intensity range ΔK

POTENTIAL VALUE DRIVER: Addresses: Documented problem 'fatigue fracture' concerning railway rail. User-submitted engineering problem (session input): "Why do rails fracture in service under fatigue loading?". [cod_optimade] retrieval status EMPTY — provider outcome recorded, not treated as absence (Art. XXI.3). [arxiv] 5 records retrieved live (2010-2020). Illustrative records: 'How we can control the crack to propagate along the specified path feasibly?'; 'Steady-state propagation of a Mode III crack in couple stress elastic materials'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: mechanical_structural

Disciplines: mechanical engineering, structural analysis, fatigue and fracture, materials science

Architecture: PROPOSED architecture (no physical system exists): load path / structural backbone; actuation or compliance element; guidance / bearing interfaces; fatigue-critical features register — realizing: apply a magnetorheological (MR) fluid coating to the rail surface and energize it with a magnetic field during service to generate a compressive stress field at potential crack tips, thereby inducing a crack tip shielding effect under cyclic loading

Subsystems:

?: load path / structural backbone - ?

?: actuation or compliance element - ?

?: guidance / bearing interfaces - ?

?: fatigue-critical features register - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: apply a magnetorheological (MR) fluid coating to the rail surface and energize it with a magnetic field during service to generate a compressive stress field at potential crack tips, thereby inducing a crack tip shielding effect under cyclic loading

Key Physics: Mechanics / structures governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Mechanics / structures governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

MEC-002: Section bending stress. Expression: $\sigma = M * c / I$. Applicability to this invention: APPLICABLE; holds under: linear-elastic material; small deflection; prismatic section. Assumptions: stress concentrators multiply local stress above the nominal value — geometry factors must be applied from the actual design. Engaged variables: σ .

MEC-003: Hooke's law (linear elasticity). Expression: $\sigma = E * \epsilon$. Applicability to this invention: APPLICABLE; holds under: below proportional limit; isothermal. Assumptions: superelastic members (e.g. nitinol) do not follow linear elasticity through transformation — use their measured plateau relations instead. Engaged variables: σ .

MEC-001: Euler buckling load. Expression: $P_{cr} = \pi^2 * E * I / (K * L)^2$. Applicability to this invention: CONDITIONAL; holds under: elastic, slender column, small deflection. Assumptions: K for catheter constraint conditions is MODEL_DERIVED. Engaged variables: NOT ESTABLISHED.

Assumptions:

- stress concentrators multiply local stress above the nominal value — geometry factors must be applied from the actual design
- superelastic members (e.g. nitinol) do not follow linear elasticity through transformation — use their measured plateau relations instead
- K for catheter constraint conditions is MODEL_DERIVED
- MEC-002 (Section bending stress) is applicable only under: linear-elastic material; small deflection; prismatic section.
- MEC-003 (Hooke's law (linear elasticity)) is applicable only under: below proportional limit; isothermal.
- MEC-001 (Euler buckling load) is applicable only under: elastic, slender column, small deflection.
- Model input variables requiring sourced values before any numeric result: M (N.m), c (m), I (m⁴), E (Pa), epsilon (-), E (Pa), I (m⁴), K (-), L (m) — all UNKNOWN (no sourced value exists; Art. VIII).
- Model output variables produced by the governing model: σ (Pa), σ (Pa), P_{cr} (N) — UNKNOWN until the inputs are sourced and the model is evaluated.

- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): member slenderness ratio [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; yield/ultimate strength [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; applied cyclic load amplitude [P_cr, N] value_status=UNKNOWN; required run-out cycles [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; articulation radius [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; friction coefficient at guides [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'fatigue fracture', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'kink/buckling below required push ability', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'fatigue crack initiation at stress concentrator', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'wear debris generation at sliding interfaces', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'joint separation under overload', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'set-release of superelastic shape after set', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	maintain structural integrity under cyclic loading	SOURCE_FACT
DI-002	Mechanism driver (proposed)	crack tip shielding effect that correlates crack growth rate with the loading from the fluid in addition to the stress intensity	MODELING
DI-003	Expected effect target	the induced compressive field reduces the effective stress intensity range almost 1/5, slowing fatigue crack propagation	MODELING
DI-004	Domain design input pattern: delivery path applicability	NOT_ESTABLISHED (candidate pattern; confirm again)	ENGINEERING_PROPOSED
DI-005	Domain design input pattern: actuation applicability	AT LEAST 100 N [force]	ENGINEERING_PROPOSED
DI-006	Domain design input pattern: duty cycle applicability	NOT_ESTABLISHED (candidate pattern; confirm again)	ENGINEERING_PROPOSED
DI-007	Regulatory design input: compliance candidate standards	ASTM F2516 (material design) pathway UNKNOWN (PERFECT PROPOSED) is never inferred	UNKNOWN
DI-008	Custodied observation: Experimental Studies / Fatigue Performance	article/1566937, content_hash 75a5a761a4d0b52680d26205a64096ba58d1d46bb5a017ab8	
DI-009	Custodied observation: Failure Analysis	https://www.fatigue/article/41157004; content_hash 97fd478765f76213769b99a67bd661ed394ef6410ca77b6	
DI-010	Custodied observation: Evaluation of Pearlitic Steel Permeability	article/41515716, content_hash 19a95427801900408a0c7ca66c917b4d32af2b1f195	
DI-011	Custodied observation: Nonlinear Ultrasonic Time-Dependent Elasticity	article/4198846, content_hash 580993883e9010a6a92a60ce5bea542d97b87e4780c9	
DI-012	Custodied observation: Fatigue Life Evaluation and Structural Optimization	article/42076298, content_hash 206263462e3905f068e9a0cd4a8a50c2b9ed77054e24c	

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	load path / structural backbone realizing apply a magnetorheological (MR) fluid coating to the tails surface and energize it with a magnetic field during service	CONCEPTUAL	details to the tails surface and energize it with a magnetic field during service
DO-002	actuation or compliance element realizing apply a magnetorheological (MR) fluid coating to the tails surface and energize it with a magnetic field during service	CONCEPTUAL	details to the tails surface and energize it with a magnetic field during service
DO-003	guidance / bearing interfaces realizing apply a magnetorheological (MR) fluid coating to the tails surface and energize it with a magnetic field during service	CONCEPTUAL	details to the tails surface and energize it with a magnetic field during service
DO-004	fatigue-critical features register realizing apply a magnetorheological (MR) fluid coating to the tails surface and energize it with a magnetic field during service	CONCEPTUAL	details to the tails surface and energize it with a magnetic field during service
DO-005	load path / structural backbone interfaces with actuation or compliance element: signal energy, artificial neural network for apply a magnetorheological (MR) fluid coating to the tails surface and energize it with a magnetic field during service	CONCEPTUAL	signal energy, artificial neural network for apply a magnetorheological (MR) fluid coating to the tails surface and energize it with a magnetic field during service
DO-006	actuation or compliance element interfaces with guidance / bearing interfaces: signal energy, artificial neural network for apply a magnetorheological (MR) fluid coating to the tails surface and energize it with a magnetic field during service	CONCEPTUAL	signal energy, artificial neural network for apply a magnetorheological (MR) fluid coating to the tails surface and energize it with a magnetic field during service
DO-007	guidance / bearing interfaces interfaces with fatigue-critical features register: signal energy, artificial neural network for apply a magnetorheological (MR) fluid coating to the tails surface and energize it with a magnetic field during service	CONCEPTUAL	signal energy, artificial neural network for apply a magnetorheological (MR) fluid coating to the tails surface and energize it with a magnetic field during service
DO-008	Device boundaries: patient/fluid inlet and outlet terminal control/data processing required by apply a magnetorheological (MR) fluid coating to the tails surface and energize it with a magnetic field during service	CONCEPTUAL	control/data processing required by apply a magnetorheological (MR) fluid coating to the tails surface and energize it with a magnetic field during service
DO-009	Flow/force/signal path geometry realizing the mechanism in apply a magnetorheological (MR) fluid coating to the tails surface and energize it with a magnetic field during service	UNKNOWN	mechanism in apply a magnetorheological (MR) fluid coating to the tails surface and energize it with a magnetic field during service
DO-010	Operating range for member slenderness ratio within which the mechanism holds its stated behavior	UNKNOWN	mechanism holds its stated behavior
DO-011	Operating range for yield/ultimate strength within which the mechanism holds its stated behavior	UNKNOWN	mechanism holds its stated behavior
DO-012	Operating range for applied cyclic load amplitude within which the mechanism holds its stated behavior	UNKNOWN	mechanism holds its stated behavior
DO-013	Operating range for required run-out cycles within which the mechanism holds its stated behavior	UNKNOWN	mechanism holds its stated behavior
DO-014	Operating range for articulation radius within which the mechanism holds its stated behavior	UNKNOWN	mechanism holds its stated behavior
DO-015	Operating range for friction coefficient at guides within which the mechanism holds its stated behavior	UNKNOWN	mechanism holds its stated behavior
DO-016	Sensing -> decision -> actuation loop maintaining the mechanism in apply a magnetorheological (MR) fluid coating to the tails surface and energize it with a magnetic field during service	PROPOSED	operating point in apply a magnetorheological (MR) fluid coating to the tails surface and energize it with a magnetic field during service
DO-017	Information path: sensing element -> decision/monitor control/data processing required by apply a magnetorheological (MR) fluid coating to the tails surface and energize it with a magnetic field during service	CONCEPTUAL	act (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical in apply a magnetorheological (MR) fluid coating to the tails surface and energize it with a magnetic field during service
DO-018	Fixture enabling: buckling / kink margin tests across duty cycle	UNKNOWN	rig concept named, build specific instrumentation selection; acceptance instrumentation
DO-019	Fixture enabling: fatigue life (run-out) testing — rig control	UNKNOWN	build specific instrumentation selection; acceptance instrumentation
DO-020	Fixture enabling: friction and wear characterization — rig control	UNKNOWN	build specific instrumentation selection; acceptance instrumentation

ID	Description	Status	Missing Inputs
DO-021	Fixture enabling: buckling/kink margin test across duty cycle	UNKNOWN	Unknown concept name; fixture specification; instrument selection; acceptance instrumentation

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
member slenderness ratio	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned and target) value and verification method k
yield/ultimate strength	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned and target) value and verification method k
applied cyclic load amplitude	UNKNOWN (no sourced value)	N	ENGINEERING_PROPOSED	(design assigned and target) value and verification method k
required run-out cycles	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned and target) value and verification method k
articulation radius	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned and target) value and verification method k
friction coefficient at guides	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned and target) value and verification method k

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
fatigue fracture	Fails by fatigue fracture: the design fails due to the stress concentration at the joint, which is not accounted for in the design.	Use of stress concentration factors and fatigue life prediction methods.	Use of stress concentration factors and fatigue life prediction methods.	Uncertainty in fatigue life prediction methods.
kink/buckling below yield	Fails by kink/buckling below yield: the design fails due to the buckling of the joint, which is not accounted for in the design.	Use of buckling analysis and design of the joint to prevent buckling.	Use of buckling analysis and design of the joint to prevent buckling.	Uncertainty in buckling analysis and design of the joint to prevent buckling.
fatigue crack initiation	Fails by fatigue crack initiation: the design fails due to the initiation of a fatigue crack, which is not accounted for in the design.	Use of fatigue crack growth analysis and design of the joint to prevent crack initiation.	Use of fatigue crack growth analysis and design of the joint to prevent crack initiation.	Uncertainty in fatigue crack growth analysis and design of the joint to prevent crack initiation.
wear debris generation	Fails by sliding and debris generation: the design fails due to the generation of debris, which is not accounted for in the design.	Use of wear analysis and design of the joint to prevent debris generation.	Use of wear analysis and design of the joint to prevent debris generation.	Uncertainty in wear analysis and design of the joint to prevent debris generation.
joint separation under load	Fails by joint separation under load: the design fails due to the separation of the joint, which is not accounted for in the design.	Use of joint separation analysis and design of the joint to prevent separation.	Use of joint separation analysis and design of the joint to prevent separation.	Uncertainty in joint separation analysis and design of the joint to prevent separation.
set-release of superelastic	Fails by set-release of superelastic: the design fails due to the set-release of the superelastic, which is not accounted for in the design.	Use of set-release analysis and design of the joint to prevent set-release.	Use of set-release analysis and design of the joint to prevent set-release.	Uncertainty in set-release analysis and design of the joint to prevent set-release.

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
fatigue fracture	Fails by fatigue fracture: the device fails by fatigue fracture	design mitigation NOT ESTABLISHED	fatigue life prediction based on data failure the people (AetX)(V)
kink/buckling below required capacity	Fails by kink/buckling below required capacity	pushability test bed depression demonstration and kink/buckling capacity test	the buckling load (AetX)(V)
fatigue crack initiation at stress concentration	Fails by fatigue crack initiation at stress concentration	stress concentration reduction	fatigue crack initiation at stress concentration (AetX)(V)
wear debris generation at sliding interfaces	Fails by wear debris generation at sliding interfaces	material pairing and coating condition	the wear debris generation (AetX)(V)
joint separation under overload	Fails by joint separation under overload	load on a larger sized area over load	the joint separation under overload (AetX)(V)
set-release of superelastic shape	Fails by set-release of superelastic shape	shape set process control	the set-release of superelastic shape (AetX)(V)

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	embed strain gauges and acoustic emission sensors	embed strain gauges and acoustic emission sensors	the sensors are not at test standing	the sensors are not at test standing
VF-002	run killer experiment: falsification_test	run killer experiment (E/C/Calc/4457363930204945) data (E/C/Calc/4457363930204945)	the test results are not as expected	the test results are not as expected
VF-003	buckling / kink margin tests across duty cycle	buckling / kink margin tests (E/C/Calc/4457363930204945) data (E/C/Calc/4457363930204945)	the test results are not as expected	the test results are not as expected
VF-004	fatigue life (run-out) testing. Acceptance criteria	fatigue life (E/C/Calc/4457363930204945) data (E/C/Calc/4457363930204945)	the test results are not as expected	the test results are not as expected
VF-005	buckling/kink margin test across the duty cycle	buckling / kink margin tests (E/C/Calc/4457363930204945) data (E/C/Calc/4457363930204945)	the test results are not as expected	the test results are not as expected
VF-006	fatigue run-out at worst-case loading	fatigue run-out at worst-case loading (E/C/Calc/4457363930204945) data (E/C/Calc/4457363930204945)	the test results are not as expected	the test results are not as expected
VF-007	buckling / kink margin tests across duty cycle	buckling / kink margin tests (E/C/Calc/4457363930204945) data (E/C/Calc/4457363930204945)	the test results are not as expected	the test results are not as expected
VF-008	overload margin demonstration to worst-case	overload margin demonstration to worst-case (E/C/Calc/4457363930204945) data (E/C/Calc/4457363930204945)	the test results are not as expected	the test results are not as expected

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification)	acceptance criteria to be fixed	validation is not performed

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	nitinol / polymer composite membrane	minimally invasive device practice	True	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
nitinol / polymer composite membrane	nitinol / polymer composite membrane	1	Material	nitinol / polymer composite membrane	NOT ESTABLISHED	NOT ESTABLISHED	Reliability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- laser-cut hypotube / braided composite shaft: domain registry manufacturing route - candidate only; no process qualification exists
- passivation and surface finish control: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-005', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter member slenderness ratio [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter yield/ultimate strength [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter applied cyclic load amplitude [P_cr N]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter required run-out cycles [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter articulation radius [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}
- {'unknown': 'value of critical parameter friction coefficient at guides [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}
- {'unknown': 'acceptance criterion for verification VF-002', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-002'}
- {'unknown': 'acceptance criterion for verification VF-003', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-003'}

- {'unknown': 'acceptance criterion for verification VF-004', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-004'}
- {'unknown': 'acceptance criterion for verification VF-005', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-005'}
- {'unknown': 'acceptance criterion for verification VF-006', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-006'}
- {'unknown': 'acceptance criterion for verification VF-007', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-007'}
- {'unknown': 'acceptance criterion for verification VF-008', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-008'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article	maintaining the proposed physical verification to be completed by 15/05/2025	embed strain gauges	NOT ESTABLISHED (constraint compliance)	NOT ESTABLISHED (constraint compliance)	NOT ESTABLISHED (to be quoted)
WP-02	constraint-compliance envelope article	per method	embed strain gauges	NOT ESTABLISHED (constraint compliance)	NOT ESTABLISHED (constraint compliance)	NOT ESTABLISHED (to be quoted)
WP-03	article for VF-001	per method	run killer experiment: NOT ESTABLISHED (VF-001 verification)	NOT ESTABLISHED (VF-001 verification)	NOT ESTABLISHED (VF-001 verification)	NOT ESTABLISHED (to be quoted)
WP-04	article for VF-002	per method	buckling / kink margin test: NOT ESTABLISHED (VF-002 verification)	NOT ESTABLISHED (VF-002 verification)	NOT ESTABLISHED (VF-002 verification)	NOT ESTABLISHED (to be quoted)
WP-05	article for VF-003	per method	fatigue life (run-out) test: NOT ESTABLISHED (VF-003 verification)	NOT ESTABLISHED (VF-003 verification)	NOT ESTABLISHED (VF-003 verification)	NOT ESTABLISHED (to be quoted)
WP-06	article for VF-004	per method	buckling/kink margin test: NOT ESTABLISHED (VF-004 verification)	NOT ESTABLISHED (VF-004 verification)	NOT ESTABLISHED (VF-004 verification)	NOT ESTABLISHED (to be quoted)
WP-07	article for VF-005	per method	fatigue run-out at work stress: NOT ESTABLISHED (VF-005 verification)	NOT ESTABLISHED (VF-005 verification)	NOT ESTABLISHED (VF-005 verification)	NOT ESTABLISHED (to be quoted)
WP-08	article for VF-006	per method	buckling / kink margin test: NOT ESTABLISHED (VF-006 verification)	NOT ESTABLISHED (VF-006 verification)	NOT ESTABLISHED (VF-006 verification)	NOT ESTABLISHED (to be quoted)
WP-09	article for VF-007	per method	fatigue run-out at work stress: NOT ESTABLISHED (VF-007 verification)	NOT ESTABLISHED (VF-007 verification)	NOT ESTABLISHED (VF-007 verification)	NOT ESTABLISHED (to be quoted)
WP-10	article for VF-008	per method	overload margin demonstration: NOT ESTABLISHED (VF-008 verification)	NOT ESTABLISHED (VF-008 verification)	NOT ESTABLISHED (VF-008 verification)	NOT ESTABLISHED (to be quoted)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 10-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.